Group Assignment Guidelines

Objective

The aim of this group assignment is to produce a report comparing the protein profiles of a microbial co-culture grown under two different conditions. **Analyse all MGF files belonging to the same sample together, and analyse each sample separately**. Identify the proteins detected in each sample, and compare their estimated relative abundances. Use biological databases to explore the functions and metabolic pathways of the identified proteins and discuss how the observed differences may relate to the growth substrate.

Data description

The dataset contains LC–MS/MS data from a syntrophic co-culture of two microorganisms: *Syntrophomonas zehnderi* (DSM 17840) and *Methanobacterium formicicum* (DSM 1535).

S. zehnderi degrades long-chain fatty acids, producing acetate and hydrogen. *M. formicicum* consumes hydrogen to reduce CO_2 to methane, helping maintain conditions that favour bacterial fatty acid degradation.

One culture was incubated with stearate, a saturated fatty acid, and the other with oleate, an unsaturated fatty acid. Each set of MGF files corresponds to one of these growth conditions (Table 1).

Table 1. Growth substrates and corresponding MGF files.

Fatty acid	Sample designation	MGF files
Oleate	Sample 1	2014_06_19_Andreia_F01.mgf to 2014_06_19_Andreia_F24.mgf
Stearate	Sample 2	2014_06_19_Andreia_F73.mgf to 2014_06_19_Andreia_F96.mgf

Report structure

Prepare a report presenting your analysis of the two samples. Organise it into the following main sections:

1. Summary

Provide a concise overview of the aim, the approach used, the main findings, and your conclusions.

2. Methods

Describe the bioinformatics workflow you followed, including the tools, databases, and relevant parameters used. Briefly explain what each step does and why it is needed.

3. Results and interpretation

Present and interpret your results, focusing on the comparison between the two samples and the functional analysis of the identified proteins. Compare protein identifications and estimated relative abundances, explore their biological functions and metabolic pathways, and discuss how the observed differences may relate to the growth substrates. Use tables and figures where appropriate, and acknowledge the limitations of your analysis.

4. Reference list

Include a reference list for the tools, databases, and literature used in your analysis. All references should also be cited in the relevant sections of the report.

Practical guidance for data analysis

These suggestions provide a starting point for the functional analysis of your data. You are encouraged to use additional tools, databases, or approaches that could enrich your analysis and strengthen your conclusions.

1. Protein identification in Galaxy

Perform the bioinformatics analysis separately for each sample, following the workflow described in the practical exercise completed in class. Use the same protein sequence database and analysis parameters for both samples.

Download the **Protein Report** and **Certificate of Analysis** for each sample. Review the identification and validation results before comparing the samples.

2. Normalisation and comparison of protein abundances

Use the **Spectrum Counting** column in the Protein Report to estimate relative protein abundances. For each sample:

- Remove contaminant entries.
- Calculate the sum of the Spectrum Counting values for all remaining protein entries.
- Divide each protein's value by this sum and multiply by 100 to express its estimated relative abundance as a percentage.

Apply this procedure independently to each sample, using the same inclusion criteria.

For each sample, calculate the denominator using the proteins from both microorganisms together. Each sample must have its own denominator.

Create a comparison table containing all protein identifiers detected in either sample, without duplicates, and the corresponding relative abundances in the two samples. You can use Excel's Remove Duplicates function, available under the Data tab, to create a list of unique protein identifiers, and XLOOKUP to match their abundances across the two samples.

These values describe each protein's relative contribution to the detected protein profile. A protein that was not detected in one sample should not automatically be considered biologically absent.

3. Retrieval of protein annotations and sequences

Use UniProt ID Mapping to retrieve information about the identified proteins. Submit the UniProt accession numbers from your reports and select **UniProtKB** as the source and target database.

Select useful annotation columns, such as protein name, organism, function, Gene Ontology terms, and EC number, where available. Download the annotation table and the corresponding protein sequences in **FASTA format**.

Organise the proteins by microorganism so that you can interpret their functions separately.

4. Assignment of KEGG Orthology identifiers

Submit the protein FASTA sequences to BlastKOALA to obtain **KEGG Orthology identifiers (KO identifiers, or K numbers)**. Analyse the proteins from each microorganism separately.

Download the results and retain the correspondence between each protein identifier and its assigned K number. Some proteins may not receive a KO assignment; this does not invalidate their identification.

5. Visualisation and functional interpretation

Use KEGG Mapper – Reconstruct to map the assigned K numbers onto KEGG pathways. Keep the results for the two microorganisms and growth conditions distinguishable.

Explore the pathways containing the identified proteins and relate them to the metabolism of each microorganism and their syntrophic interaction. Combine the pathway information with your abundance comparison to discuss potential differences between the stearate and oleate conditions.

The detection of a protein associated with a pathway does not demonstrate that the entire pathway was detected or establish its metabolic activity.